



LT1, LT2 and VO2 for Running

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Overview

LT1, LT2, and VO2 are the three reference points that anchor running zones, test results, and race pacing. The physiology is the same in every sport, but running has its own quirks: it uses the most muscle mass of the three disciplines, it spikes heart rate more than swimming or cycling, and impact forces add a mechanical limit that does not exist on the bike or in the pool.

In running, LT2 is called lactate threshold or anaerobic threshold, and it shows up as threshold pace. This guide covers what each marker means on the run, how to find it, and how to train around it.

LT1 for running

LT1 is where lactate first builds above baseline, often between 1.5 and 2.5 mmol/L. It sets the upper boundary of true aerobic base running. Training below it builds fat use, blood volume, oxygen carrying capacity, and bone, ligament, and tendon strength with little fatigue.

LT1 is hard to find without lactate sampling or gas analysis. A practical approach is to find LT2 first, then calculate LT1 as a percentage of it:

Athlete category	Typical profile	LT1 as a percentage of LT2
Age-Group	Newer athletes, about 6 to 10 hours per week	About 70%
Performance	High-performance age groupers	About 75%
Elite	Professional athletes (testing is typically done)	About 80%

The Zone 2 running problem. Running spikes heart rate more than swimming or cycling, and heart rate is also more variable at low intensities because of hydration, temperature, stress, and the autonomic nervous system. Very few athletes can run continuously at true Zone 2. Three fixes work:

- **Run at the slowest sustainable pace at which running biomechanics are not negatively affected.** Do not force a pace so slow that form breaks down. The goal is the lowest intensity you can hold with good mechanics.
- **Walk-run protocols.** For example, 3 minutes running and 3 minutes walking. The walking intervals let the athlete recover so the running portions stay at the intended low intensity. Speed walking also works.
- **Let swimming and cycling build the base.** Triathletes often run better than single-sport runners because swim and bike build the aerobic base without extra running volume and injury risk.

LT2 for running

LT2 is the tipping point where the body accumulates more lactate than it can clear, typically around 3.8 to 4.1 mmol/L. It is the primary reference for tempo and threshold work, and most athletes can hold it for roughly 30 to 60 minutes in race conditions.

You can find it in the field without a lactate analyzer:

- **Breathing.** At threshold, breathing is labored but still controlled. Below it, breathing is comfortable and conversational. Above it, breathing turns ragged.
- **Talking.** Inability to speak in full sentences points to LT2 or above.
- **RPE.** About 7 out of 10, or 13 to 14 on the Borg 6 to 20 scale. Hard but sustainable.
- **Heart rate deflection.** At LT2 the rise in heart rate slows even as pace keeps increasing. Plot heart rate against pace from a hard progressive effort or race and look for the bend.
- **Changes at familiar paces.** If an athlete who held threshold at 6:00 per mile with controlled breathing now shows threshold breathing at 5:45, fitness has moved and it is time to retest.

VO2 for running

VO2 max is the maximum rate of oxygen use. VO2 peak is the highest value seen in a test that did not meet the strict plateau criteria. Most field tests give you VO2 peak, and for setting zones and tracking progress that is usually enough.

- **Lactate at VO2 effort:** typically 8 to 15 mmol/L or more, depending on anaerobic capacity.
- **Where it matters:** racing under an hour, particularly short-course athletes. The 800 m test is a good window into speed endurance and VO2 capability.
- **Training it:** high-intensity intervals at roughly 95 to 105% of threshold.
- **Its ceiling:** after the first two to three years of intense training, gains come more from threshold and running economy than from further VO2 increases.

Running economy, the energy cost of holding a given pace, is shaped by biomechanics, muscle efficiency, and neuromuscular coordination. Two runners with the same VO2 can race very differently because of it.

Testing LT1, LT2, and VO2 on the run

Track testing uses progressive stages of 800 to 1600 m at increasing paces, with brief stops for blood sampling and lap times recorded for exact pace. Treadmill testing offers precise speed control and easier sampling, but may not fully replicate outdoor running biomechanics. A standard protocol uses 3 to 5 minute stages starting well below threshold and progressing to near-maximal effort. Keep running form consistent through the test and watch for the breathing changes that signal threshold.

Useful field tests (make sure athletes have been signed off by their doctor for physical exercise, and only complete maximal tests once the athlete is well conditioned to training) include the 800 m and 1600 m pair for regression rate, and the 4 x 1600 m test with 90 seconds of recovery, which predicts 5K pace.

Running power is an estimate, not a direct measurement. It is useful for pacing and for tracking an individual athlete over time, but running power and cycling power are not on the same scale. Run thresholds and zones need their own testing.

Retest every 3 to 6 months, and test in the first week after an adaptation week when the athlete is fresh.